

HDI Predictor Documentation

Performance Testing

Date	05 July 2025
Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache Bench (ab) and manual Postman requests
Type of Testing	Load Testing, Response-Time Testing
Target Module / API	/predict endpoint (Flask)
Test Environment	Local (Dockerized container on developer machine)
Test Date	05 July 2025

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single prediction request with valid indicator values	1	5	Returns correct tier and score in < 1s
2	Repeated requests to check model reload overhead	10 sequential	15	Model loads once at startup; subsequent calls stay fast
3	Concurrent requests to /predict	20 concurrent	20	No errors; consistent response times
4	Invalid/missing input fields	5	5	Returns a graceful validation error, not a server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~0.18 seconds	Pass	RandomForest inference on 4 features is very fast
2	Response Time (Max)	< 5 seconds	~0.6 seconds	Pass	Observed under 20 concurrent

					requests
3	Throughput (Req/sec)	—	~45 req/sec	Pass	Sufficient for expected single- user/demo usage
4	Error Rate	< 1%	0%	Pass	No failed requests during testing
5	CPU Utilization	< 80%	~35%	Pass	Lightweight model, no GPU required
6	Memory Utilization	< 80%	~40%	Pass	Model artifact is small (RandomForest, tabular features)

Step 4: Observations & Analysis

Because the model operates on only four numeric features with a modest RandomForest ensemble, inference is extremely fast and CPU-light. The main latency contributor is Flask's request/response cycle rather than model inference itself. No performance bottlenecks were observed at the tested concurrency levels; the primary risk for future scaling would be running many concurrent users behind a single-process Flask dev server, which is mitigated in production by running behind Gunicorn/Docker with multiple workers.

Step 5: Screenshots / Evidence

Apache Bench summary output and Postman response-time screenshots to be attached alongside this document in the submission package.